

Barei evolution algebra audit confirmation

The principal mathematics survived the bounded audit under the stated hypotheses. No fatal mathematical error was found in the reconstructed principal claims. All eleven exact check groups also passed on a fresh rerun.

Subject: Andres Barei’s On Solvable Evolution Algebras and a Conjecture by García-Martínez and Pérez-Rodríguez. The manuscript credits Muse Spark and names Nicolás Jaramillo Torres as reviewer. This separate audit was commissioned by **Mark S. Weinstein** and prepared with AI assistance.

Prepared 3 October 2026. This companion preserves the earlier detailed audit and records a follow-up in the same assistant workflow. It is source-informed, not blind replication, an independent human expert review, or a proof-assistant formalization. The material remains a draft for review; nothing has been published or sent to the authors.

Principal claim	Verdict	Scope / value of the audit
Example 2.1	Supported	Nonsolvable over $\mathbb{C}$; zero is the only complex idempotent. Complete two-branch elimination supplied.
Lemma 3.1	Supported	Greatest perfect subspace follows from descent, stabilization and inclusion.
Theorem 3.2	Supported	Elementary stable-series characterization; valid for general finite-dimensional algebras.
Theorem 3.3	Supported	A precise older existence theorem supplies a concise alternative to the compressed geometric step.
Corollary 3.4	Supported	Apply the existence result directly to the stable term. Its sufficient condition is strictly broader than Hu-Wen’s.
Remarks 3.5–3.6	Qualified	3.6 is supported. 3.5 is a search observation, not a proved universal statement.

What the audit contributes

It supplies a readable counterexample calculation, simpler proof routes, an explicit isomorphism with Hu-Wen’s exceptional example, and a four-dimensional witness proving strictness of the sufficient-condition comparison. The witness was independently constructed in the earlier audit and rechecked here; its historical novelty has not been established.

Assumptions and credit

The abstract claims concern finite-dimensional evolution algebras over an algebraically closed field; Example 2.1 is over $\mathbb{C}$. Multiplication is bilinear and generally nonassociative. The derived term is the span of all products in the preceding term. Natural-basis mixed products are zero.

Credit: Barei; named paper reviewer Nicolás Jaramillo Torres; Muse Spark as attributed by the authors; original conjecture authors Xabier García-Martínez and Andrés Pérez-Rodríguez; concurrent authors Xing-Yu Hu and Ran Wen; and Yuri Lyubich and Alexander Tsukerman for the older existence theorem. These credits imply no endorsement of this audit.

The calculation and added witness

An exhaustive complex-idempotent calculation

Rows of Barei's matrix encode e_i^2 ; mixed natural-basis products vanish. Let $r_1 = (-2, 0, 2)$, $r_2 = (0, 1, -1)$. Then $r_1^2 = 4(r_1 - r_2)$, $r_1 r_2 = 2r_2$, and $r_2^2 = 0$. Hence the derived dimensions are $3 \rightarrow 2 \rightarrow 2$: the algebra is nonsolvable.

$$x = -2x^2, \quad y = y^2 - z^2, \quad z = 2x^2 - y^2 + z^2.$$

The first equation forces $x = 0$ or $-1/2$. Put $s = y + z = 2x^2$. The other equations force $(1 - 2s)y = -s^2$. For $x = 0$, $y = z = 0$. For $x = -1/2$, $0 = -1/4$ is impossible. These two branches exhaust every complex solution; the code evaluates their exact coefficients.

A shorter route through the stable term

Descent and finite-dimensional stabilization give $D^2 = D$. Every V with $V^2 = V$ lies in every derived term, hence in D . This proves Lemma 3.1 and makes Theorem 3.2 immediate. No evolution-algebra structure is needed for these stable-series facts.

Lyubich-Tsukerman's Theorem 2.1 states the idempotent-or-square-zero existence property for finite-dimensional algebras over any algebraically closed field. A nonzero subalgebra with $N(V) = \{0\}$ therefore contains an idempotent. This proves the substantive converse of Theorem 3.3; applying it to D directly proves Corollary 3.4. Its classical geometric proof was reviewed separately from the executable checks.

Strictness witness over $\mathbb{C}$

Four-dimensional W	1	0	-1	0
	1	1	0	1
	0	1	-1	-1
	0	0	1	1

This natural-basis matrix has stable term $D = \{(x, y, z, x+z)\}$ and derived dimensions $4 \rightarrow 3 \rightarrow 3$. In D coordinates:

$$(x, y, z)(u, v, w) = (xu + yv, yv + zw, xw + zu).$$

A square-zero vector satisfies $x^2 + y^2 = y^2 + z^2 = 2xz = 0$. Since $xz = 0$, either branch forces $x = y = z = 0$. Thus $N(D) = \{0\}$.

The three coordinate bilinear forms have matrices $B_1 = \text{diag}(1, 1, 0)$, $B_2 = \text{diag}(0, 1, 1)$, and B_3 with ones at $(1, 3), (3, 1)$. Let $B_0 = B_1 + B_2$. A natural basis would simultaneously diagonalize the forms by congruence, forcing $B_0^{-1}B_1$ and $B_0^{-1}B_3$ to commute. Their commutator has entries 1 at $(1, 3)$ and -1 at $(3, 1)$, so no natural basis exists.

Consequently this evolution algebra satisfies Barei's $N(D) = \{0\}$ condition while failing Hu-Wen's condition that D be an evolution algebra. It proves strict enlargement of the sufficient hypothesis. Exact code checks the embedding, product identities, commutator and an idempotent; the quantified arguments above are separately reasoned.

Claims mapped to evidence

Primary locations use printed PDF pages. “Audit” refers to the unchanged twelve-page detailed report. Every conclusion below was revisited in this follow-up; the original construction and proofs are carried forward rather than presented as new discoveries. CLAIM_EVIDENCE.csv supplies the full fourteen-row mapping.

Publication statement	Exact source location	Evidence and limit
Counterexample survives	Barei Example 2.1, pp. 2–3; GP Conjecture 3.6, p. 602 / PDF p. 6	Audit p. 3; C01–C03 plus exhaustive branch argument. Matrix identities alone do not establish solution completeness.
Stable-series characterizations survive	Barei §2, p. 2; Lemma 3.1 and Theorem 3.2, p. 3	Audit p. 4; induction, stabilization and greatest-subspace proof. C02 illustrates the result for this example.
Theorem 3.3 survives with explicit support	Barei p. 3; LT Theorem 2.1, p. 2; Lemma 3.2, pp. 4–5; proof p. 5	Audit p. 5; homogeneous quadrics, simple trivial point, proper finite-intersection Bézout and scaling. No executable proof of the geometric theorem.
Corollary 3.4 follows directly	Barei pp. 3–4	Audit p. 6; apply the existence lemma to D itself. Finite dimension and field hypotheses retained.
Hu-Wen hypothesis implies Barei’s	Barei Remark 3.6, p. 4; HW Proposition 2.8, p. 6	Audit p. 6; independence of the natural-basis squares in perfect D forces $N(D) = \{0\}$.
Counterexamples have the same class	Barei Example 2.1; HW Theorem 2.1 / Corollary 2.2, p. 3; exceptional matrix p. 4	Audit p. 7; C04 verifies $e_1 \mapsto -h_2$, $e_2 \mapsto h_3$, $e_3 \mapsto -h_1$ and all nine products. No discovery-provenance inference.
The sufficient condition is strictly broader	Audit construction compared with Barei Corollary 3.4 and HW Proposition 2.8	Audit p. 8; C05–C06 plus square-zero completeness and congruence obstruction. No historical novelty claim for the witness.
AI collaboration is author-reported	Barei title / AI-use statement, p. 1; Meta announcement, sixth-paper section	Audit p. 10; manuscript labels and official account. Interaction records are absent; causal model contribution is not independently reconstructed.
Novelty and open questions remain bounded	HW counterexample p. 3; LT existence theorem p. 2; Barei Remark 3.5 p. 4	Audit pp. 6–7,10,12; named sources establish overlap and the older ingredient. Global priority and the search question remain unresolved.

The geometric dependency is disclosed

For a nonzero n -dimensional subalgebra, consider $Q_i(x) - tx_j = 0$ in projective n -space. The point $(0:1)$ has Jacobian $-I$ in the $t = 1$ chart and multiplicity one. An infinite solution set has another point; a finite proper intersection has degree $2^n > 1$. A nontrivial point gives a square-zero element or, after scaling by t , an idempotent. Algebraic closure supplies solutions in the ground field. Vakil’s exercise 1.8 (p. 4) and §1.10 (p. 6) support the qualified degree product.

The exact Harris fixed-point page cited generally by Barei remains unresolved; the needed existence result is supported by LT. Direct verification of the older 2017 classification row, first-discovery chronology and model-interaction provenance also remain unresolved.

Actual results and how to inspect them

Fresh rerun: Python 3.12.14; exit code 0; **11 of 11 exact groups PASS**. Recorded at 2026-10-03 15:04:32 UTC / 11:04:32 a.m. EDT. The unchanged checker's evidence records match the earlier results. Five freshly downloaded central primary PDFs match the earlier source hashes.

Run from the package directory without Python optimization:
python3 verify_audit.py --output my_results.json

Groups	Finite computation / branch-check scope	Result
C01-C03	Counterexample products, derived dimensions [3,2,2], coefficients in all analytically forced idempotent branches.	PASS
C04	Invertible signed permutation; nine basis-product identities with Hu-Wen E_2 .	PASS
C05-C06	Four-dimensional witness embedding, nine identities, dimensions [4,3,3], nonzero commutator and one idempotent.	PASS
C07	All nine vectors in the F_3 field-hypothesis example: only zero is square-zero or idempotent.	PASS
C08	Barei's numeric example reduces in characteristic two to dimensions [3,1,0,0], hence is solvable.	PASS
C09	A perfect dual-number stable term shows the sufficient condition is not necessary for equivalence in each algebra.	PASS
C10	Deleting the $+2e_3$ coefficient introduces the idempotent $-e_1/2$; the negative control detects it.	PASS
C11	A square-zero-free line lacks multiplication closure; arbitrary vector subspaces do not suffice.	PASS

Correctness, reproducibility and integrity

The mathematical conclusions rest on the stated calculations and proof dependencies. Reproducibility is supported by the unchanged program and actual outputs. File integrity is supported by SHA-256 comparison with trusted reference bytes. The original archive's seven listed file hashes and ZIP check pass; matching hashes do not prove mathematics or authorship.

The full claim table, exact outputs, run log, source records and preserved earlier artifacts accompany this PDF. The geometric existence result, infinite-dimensional argument, witness square-zero proof and implication from a natural basis to commuting operators are reasoning checks, not machine-verified universal theorems. No Grounded DI or Protocol A causal benefit is claimed.

Primary sources and credit links

Barei / Meta manuscript and original attribution

García-Martínez and Pérez-Rodríguez / original conjecture

Hu and Wen / concurrent counterexample and condition

Lyubich and Tsukerman / 2014 existence theorem

Vakil / properly qualified Bézout notes

Meta / public account of the collaboration